# Lab 1: Implementation of a Simple Reflex Agent and Model Based Agent using Vacuum Cleaner Problem

## Objective

- Understand the concept of a Simple Reflex Agent and Model Based Agent
- Simulate an AI agent environment in python
- Implement condition-based decision making
- Understand limitations of agents without memory

## Problem

Using a simple reflex agent

In the Vacuum Cleaner Problem:

- There are two rooms: **Room A** and **Room B**
- Each room can be either:
  - Dirty
  - Clean
- The agent can:
  - Clean the current room
  - Move to another room

## Working Principle

The agent follows simple rules:

| Condition | Action |
|---|---|
| Current room is Dirty | Clean the room |
| Current room is Clean | Move to another room |

Since the agent has **no memory**, it continues moving between rooms even after both rooms are cleaned.

## Algorithm

1) Initialize Room A and Room B as Dirty
2) Choose a random starting room
3) Repeat for a fixed number of steps:

- Check current room status
- If room is dirty:
  - Clean the room
- Else:
  - Move to another room

4) Display room status after every step
5) Stop the program

## Pseudocode

```
START

Initialize Room A as Dirty
Initialize Room B as Dirty

Choose random starting room

REPEAT for fixed number of steps

   IF current room is Dirty THEN
      Clean the room
   ELSE
      Move to another room
   END IF

   Switch room

END REPEAT

STOP
```

****************************************************************************************************

## Pseudocode for model based agent

```
START

Initialize Room A as Dirty
Initialize Room B as Dirty

Initialize Memory for Room A as False
Initialize Memory for Room B as False
```

```
Choose a random starting room

REPEAT

   Display current room

   IF current room is Dirty THEN
      Clean the room
      Mark current room as cleaned in memory
   ELSE
      Display "Room already clean"
   END IF

   Display room states
   Display memory states

   IF all rooms are marked cleaned in memory THEN
      Display "All rooms are cleaned"
      STOP
   END IF

   IF current room is A THEN
      Move to Room B
   ELSE
      Move to Room A
   END IF

END REPEAT

STOP
```